



MedSAM Segment Anything in Medical Images

A source-faithful paper talk from Ma et al. 2024

Research Presentation Program

Department of Statistics & Data Science

2026-08-29



香港中文大學
The Chinese University of Hong Kong

The evaluation object is overlap under a prompted mask

$$\text{DSC}(G, S) = \frac{2|G \cap S|}{|G| + |S|}$$

Here G is the expert annotation and S is the predicted segmentation mask; the paper reports DSC for internal and external validation tasks.

Model components

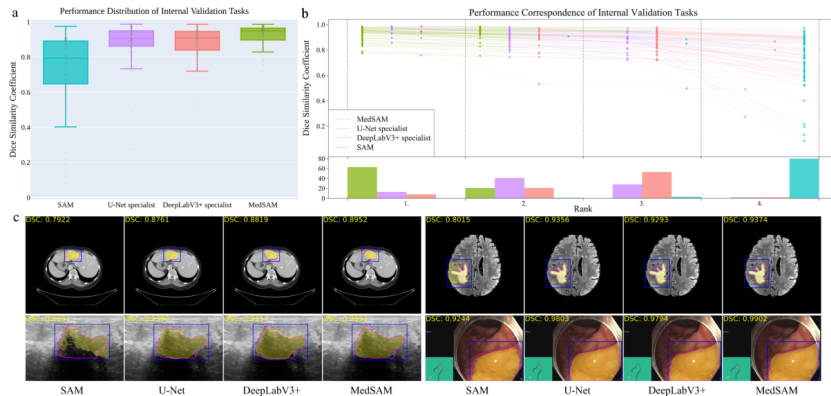
- ground-truth mask G
- segmentation mask S
- Dice similarity coefficient

Interpretation

MedSAM is judged by segmentation overlap after a bounding-box prompt specifies the target region.

Source: Ma et al. 2024 Methods, DOI 10.1038/s41467-024-44824-z.

Internal validation pairs task distributions with real qualitative cases



The source figure shows both task-level DSC/rank behavior and real article medical-image examples.

Figure 3, Ma et al. 2024, CC BY 4.0.

MedSAM keeps SAM's prompt path but trains it on medical masks

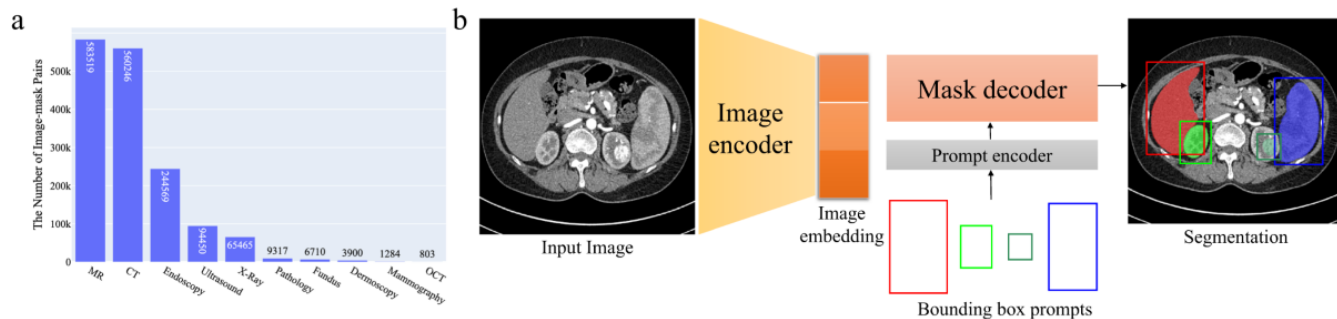


Figure 2. Medical mask distribution to the promptable SAM-style architecture used for medical segmentation.

The paper's claim is broad utility, with validation boundaries

Failure evidence

CT, MRI, and endoscopy dominate the training set, so modality imbalance remains a source-discussed limitation.



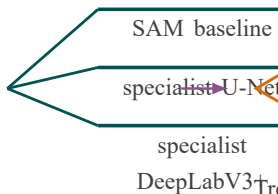
Training diversity

motivates

Manipulate sampling

- inspect unseen targets
- preserve prompt semantics
- validate by modality

Comparator arms



Decision rule

go/no-go

Treat MedSAM as a broad promptable foundation model whose deployment claim still depends on target-specific validation.

The deck ends with the article's balance: broad performance evidence plus explicit validation boundaries.

The external CT case is a same-patient segmentation comparison

